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AE 4803 – FTE

March 18th, 2026

Assignment 5 – Flight Test Reports

Flight Test Report and Evaluation for Cessna 182x

Flight Summary

The flight test was conducted by Patrick Lister, Ajay Bardakjy and I in a Cessna 182x on February 7th, 2026, with instructor pilot Jeff Perry. We departed around 0830 and returned at approximately 0930. The objective of the flight was to collect performance and flight quality data for several maneuvers including climb performance, cruise air speed evaluation, level acceleration, turn performance, and other observations.

It was a beautiful day at PDK. Weather conditions were favorable with clear skies other than reported winds at about 310° at 11 kts with gusts up to 18 kts. Visibility was 10 miles and the altimeter setting was 30.10. Aloft winds were about 343° at 41 kts at 4000 ft in elevation and a temperature of about 1° C. These conditions introduced a bumpy ride with moderate wind effects but did not significantly impact the ability to perform the test.

The aircraft configuration for most maneuvers consisted of flaps up with power set as required, which was consistent with the card instructions. The following test points were flown:

- Max power takeoff
- Max rate climb performance
- Cruise air speed at 4000 ft pressure altitude
- Level acceleration
- Turn performance
- Flight quality investigations which included phugoid and Dutch roll behaviors
- Stall maneuvers

Changes to the Airspace plan & flight profile & why

Only minor adjustments were required to the planned flight profile. Winds in the operating area were stronger than expected, which required slightly longer stabilization times during some maneuver setups. These winds really just influenced heading control and required a bit of extra attention to maintain altitude tolerances during turns and accelerations.

The pilot recommended against performing a saw tooth climb as well as changing the maximum climb rate to 2000-4000ft in order to allow for more time.

Maneuver sequence and setups comments

The test was performed per the test plan, and the climb and cruise tests were done first. During the level acceleration, the aircraft accelerated from about 65 KIAS to 140KIAS in about 60 seconds while maintaining near-constant altitude around 4000ft.

Turn performance maneuvers were conducted at about a 45° bank angle. This was generally stable, but the wind effects required small corrections to be made. The Dutch roll and phugoid behaviors were conducted at about 80 KIAS so that the oscillations could be observed.

Quality assessment of data

The overall quality of the data was acceptable for this report. The original flight test sheet can be viewed in figure 1.

Altitude tolerances during the level acceleration remained within +/- 100ft. Airspeed during cruise tests remained within the expected variability range, with speeds generally between 123 and 129 KIAS.

The turn performance had minor altitude variations due to wind disturbances but still stayed within acceptable tolerances. Overall, the data that was collected provides a reliable quick evaluation of the aircraft's performance.

Key data captured

This section will outline the crucial plots and tables used

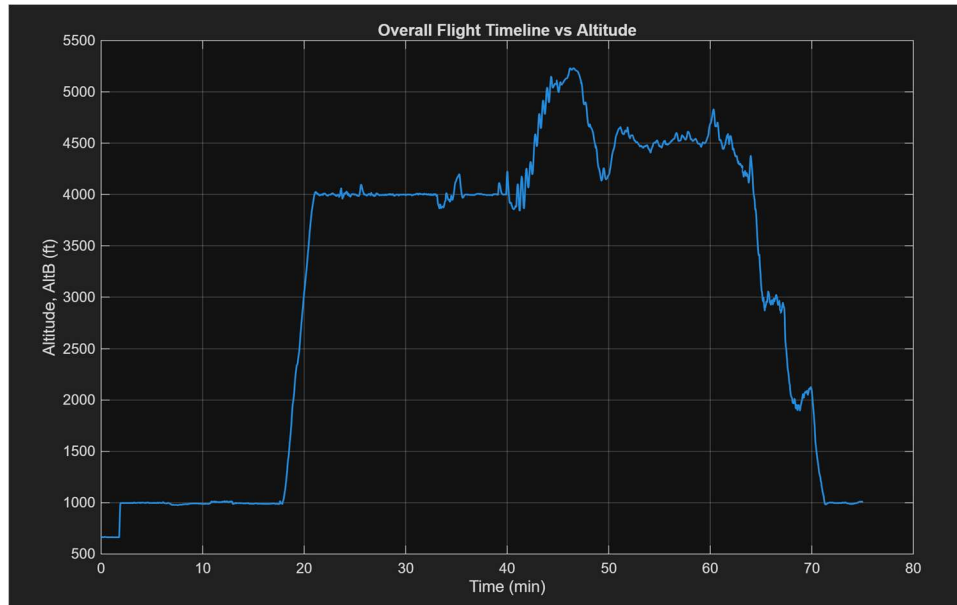


Figure 2. Overall Flight Timeline vs Altitude

Table 1. Cruise Airspeeds Table:

Time (s)	Speed (kt)	Fuel Flow (gph)	OAT (C)
0	128	16.8	-1
10	129	16.8	-1
20	129	16.8	-1
30	123	16.8	-1
40	124	16.8	-1
50	127	16.8	-1
60	125	16.8	-1

Table 2. Level Acceleration Table:

Time(s)	Speed (kt)	Altitude (ft)	Fuel Flow (gph)
0	65	4060	5
10	85	4100	21.1
20	101	4010	21
30	125	4000	21
40	136	4000	21.1
50	138	4000	21.1
60	140	4000	21

Recorded Level Acceleration:

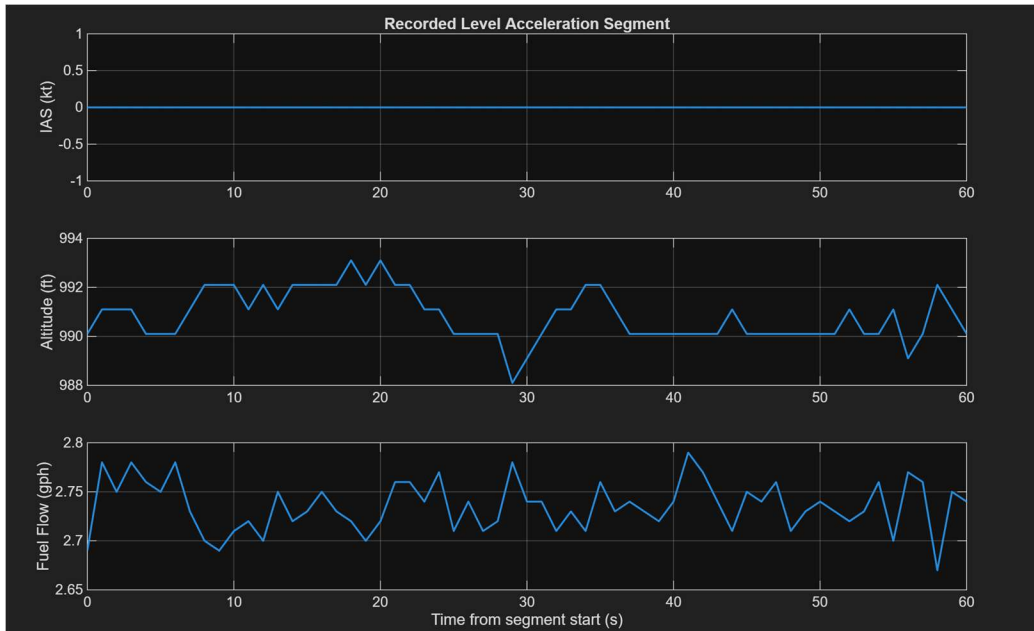


Figure 3. Recorded Level Acceleration

Table 3. Turn Performance Table:

Turn Direction	Turn Time	Start speed (kt)	End speed (kt)	Start altitude (ft)	End altitude (ft)	Change in altitude (ft)
Right	31	101	97	3910	3960	50
Left	38.3	92	99	4000	3850	-150

Table 4. Maximum Rate Climb Table:

Pressure Altitude (ft)	OAT (C)	Fuel Flow (gph)
2000	2	21
2500	2	21
3000	1	21
3500	0	20.8

Table 5. Stall Summary Table:

Stall Number	Event time (s)	Warning speed (kt)	Stall speed (kt)	Attitude at stall (ft)	G break
1	1524	74.87	65.01	3986.1	-.18
2	1615	138.03	129.03	3985.1	-.23
3	1692	118.44	109.19	3997.1	-.09

Flight Overview plots:

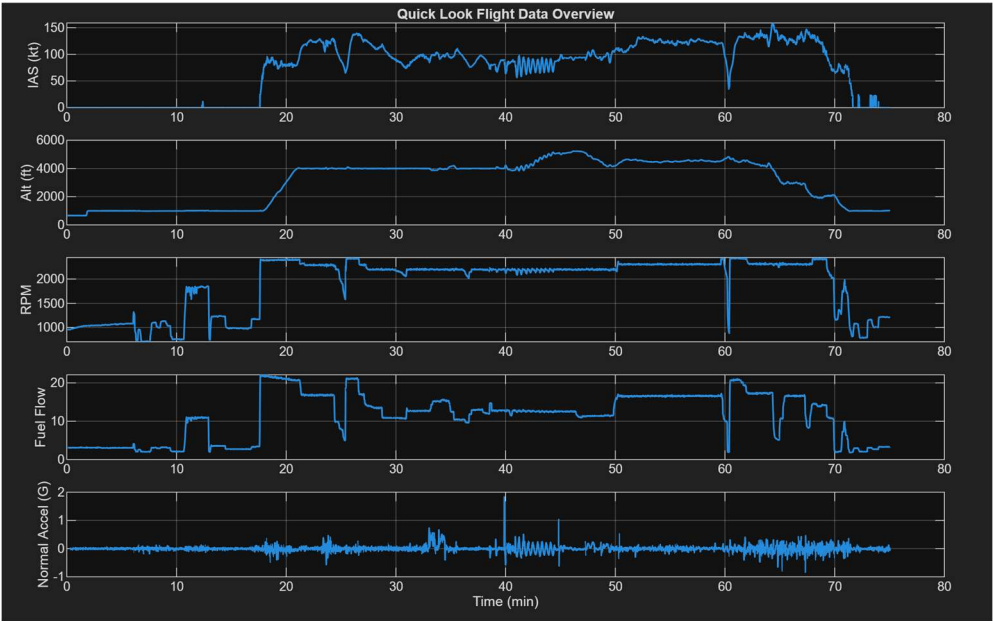


Figure 4. Quick Look Flight Data Overview

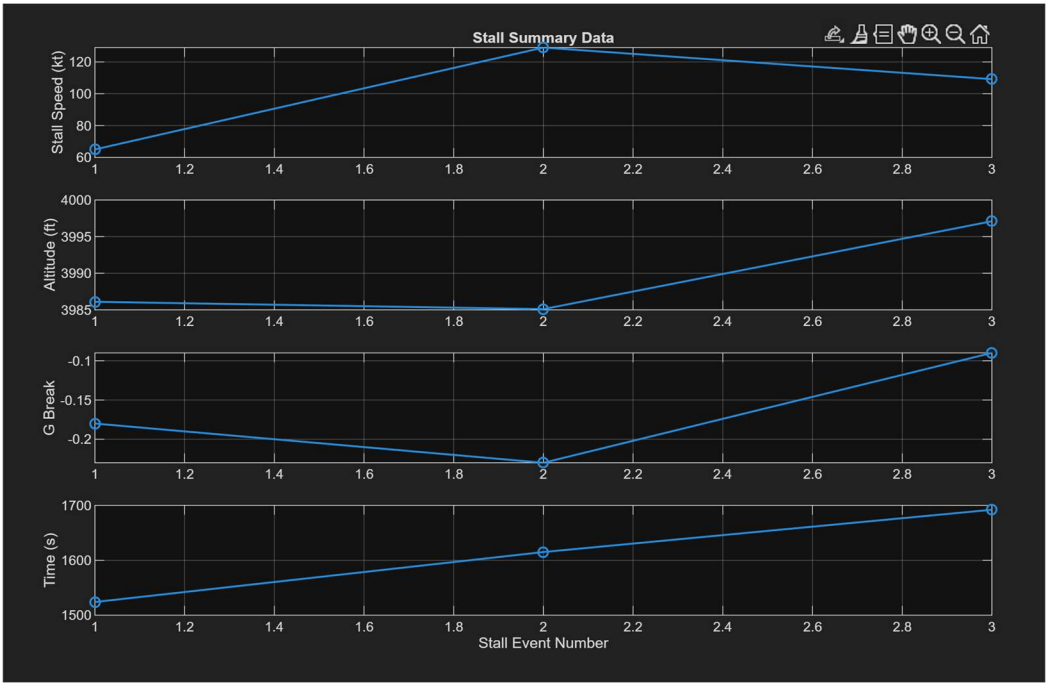


Figure 5. Stall Summary Data

Recommendations and lessons learned

Overall, the flight provided useful data for the required maneuvers, but there are several ways the testing process could be improved in future flights. One recommendation is to allow more time to stabilize the aircraft before starting each maneuver. In some cases, wind conditions made it harder to maintain the desired altitude and air speed, which could affect data quality. Taking additional time to trim the aircraft and reaching steady conditions would help keep the flight parameters within the specified data bands.

Another improvement would be better coordination between maneuver timing and data recording. Clearly noting the start and end of each maneuver during the flight would make it easier to match the recorded flight data with the observations written on the test cards.

This flight also showed how environmental conditions such as wind can affect maneuver execution and data consistency. Being aware of these effects and making small corrections during the setup phase can improve the overall quality of the data collected.

Overall, the flight was successful in collecting the necessary performance and stall data, and the quick-look analysis helped confirm that most maneuvers were completed within acceptable tolerances. The experience also helped reinforce the importance of careful maneuver setup and clear data recording during flight testing.

Figures and Data:

C-172X/NXXXX	LIMITED FLIGHT EVAL		Date: 7/2/26	
TC: Patrick Lister	IP:			
Step:	ETD/ATD: 8:39	ETA/ATA: 9:30		
Frequencies		Nav aids		
PDK GND	121.6	PDK VOR	101.6	
PDK TWR	120.9	PDK ILS	111.1	
ATIS	128.4			
AWOS	128.4			
ATL CONTRL	126.97			
Weather				
PDK	4' 310 ll - 5 dew aft			
	vis 10 gust 18			
ATL	310 at 12 gusting 21 vis 10 +			
	sky clear. 3°C dew - 5 alt 30.			
ALOFT	winds 343 at 41 - 1°C 4000 ft			
NOTAMS				
PDK	31 right out			
Obbs	Tach	IP Initials	Fuel	Landings
6706.3	5846.4			
				A

C-172X	LIMITED FLIGHT EVAL		Date:
Event	Objective		
Ground Block / TO	Spot Check		
Climb / Cruise	Spot Check		
LA / Turn / Stall	Collect Data		
FQ Observations	Observe, Collect Data		
Descent / Landing	Spot Check		

Event	Conditions/Config	Data Required
1. Takeoff	Flaps Up	Not Required
2. Max Rate Climb	Flaps Up, 1000-3000 Ft PA	Time, Fuel Flow, OAT
3. Cruise Airspeeds (Range/Endurance)	Flaps Up, 4000 Ft PA	Fuel Flow, RPM, OAT
4 Level Accel	4,000ft PA	Time, V
5. Cruise	3-5000ft PA	V, FF, OAT, Time
6. Turn Perf	2,000ft PA	V, Bank, time
7. FQ Observations	80 KIAS, >2000 Ft	Handheld data and observations
8. Stall Investigation	Light Weight, Cruise, Flaps Up >1000 Ft PA	Stall warning speed, V _{SO} , V _{S1} , Pilot Comments
9. Landing	As Required	Distance

B

Maximum Power Takeoff			1
Configuration	Limits		
Flaps 15 10°	V _I = 85 KIAS		
Pwr As Req.			
OAT <u>5</u>	Time _____	Fuel <u>0.1</u> gal	
Winds <u>029 / 7 kt</u>		Distance <u>1.0</u> ft	
Comments:			
Maximum Rate Climb			2
Configuration	Limits	Data Band	
Flaps Up	N/A	2100-3000 Ft PA	
Pwr Maximum			
A/S per Flt Man			
Time	PA	OAT	Fuel Flow
0	500		"
	1000		
	1500		
0	2000	2°	21.2
	2500	2°	21.0
23	3000	1°	21.0
	3500	0°	20.8
		-1°	20.6

[illegible]

Level Accel 4000ft PA				4
Configuration		Limits	Data Band 3-5K	
Flaps Up		V _{limb} = 105 KIAS	Tolerances: +/- 100ft	
Pwr As Req.			+/- 2 KIAS	
On conditions, trim shot, with full power hold altitude with forward pressure and accelerate from 1.2V _{stall} to V _{lim}				
Notes:				
Time	Airspeed	Altitude	Fuel Flow	
0	65	4000	5.0	
10	85	4100	21.1	
20	101	4000	21.0	
30	125	4000	21.0	
40	136	4000	21.1	
50	138	4000	21.1	
60	146	4000	21.6	
Comments:				

Saw Tooth Climbs					5
Configuration		Limits	Data Band		
Flaps Up		V _I = 85 KIAS	0-100 Ft AGL		
Pwr As Req.					
Procedure:					
OAT	Winds				
Comments:					

Turn Perf 2000ft PA			6
Configuration		Limits	Data Band
Flaps Up		V _{cruise} KIAS	+/- 200ft
Pwr As Req.			
Task: Time to complete 360deg turn at 45deg bank angle. Note airspeed and altitude changes			
START	Right	Left	
Time:	0	0	
Airspeed:	101	92	
Altitude:	3440	4000	
END	315	38.315	
Time:			
Airspeed:	97	99	
Altitude:	3460	3500	
Complete on frontside & backside airspeeds			

FQ Investigations			7
Configuration		Limits	Data Band
Flaps Up		V _I = 85 KIAS	V _I = 80 KIAS +/- 5
Pwr As Req.		H _I > 5000 Ft PA	
Phugoid	18 # Overshoots	Time	00 28
		V	85
		h	
Short Period	# Overshoots	Time	
		V	
		h	
Dutch Roll	3 # Overshoots	Time	04 28
		V	94
		h	
Spiral Stab	# Overshoots	Time	01 28
		V	91
		h	
Comments: Control inputs?			

Stall Investigation		8
Configuration	Limits	Safety
Flaps Up Pwr As Req.	1.3V _{stall} = KIAS 1.2 1.1 H>1000 Ft PA	Recovery Procedures
Clean	Warning Speed	50
Flaps 10 10°	Stall Speed	40
PA Config	Warning Speed	
Flaps 30 40°	Stall Speed	
Comments/Characteristics:		
REPEAT POWER ON/OFF		

Figure 1. Original Data Sheet